

PLASTIC-CRAFT

Polystyrene

GPPS (General Purpose) & HIPS (High Impact) Comparison

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The most economical clear (GPPS) and impact-resistant (HIPS) thermoplastics available. GPPS offers crystal clarity and high stiffness; HIPS adds rubber-modified toughness for structural applications. Both are easy to fabricate, paint, and print.

STOCK AVAILABILITY

Plastic-Craft maintains extensive inventory across a wide range of standard and specialty sizes in sheets, rods, tubes, and custom profiles. Most orders ship from stock — contact us for immediate availability.

WHEN TO CHOOSE

GPPS Glazing, displays, light diffusers, optical lenses, lab ware

HIPS Housings, packaging, signage, thermoformed trays, prototypes

KEY HIGHLIGHTS

Lowest cost per pound
GPPS: 88% light transmission
Excellent printability
Easy to thermoform
FDA compliant grades
Solvent-bondable

✓ **Most cost-effective plastic for displays & packaging**

COMMON APPLICATIONS

Point-of-purchase displays
Light diffusers & lenses
Food packaging & trays
Appliance housings
Model & prototype parts
Picture frames & glazing
Electronics enclosures

GRADES AVAILABLE

GPPS (crystal clear)
HIPS (opaque white/colors)
FDA-compliant
UV-stabilized
Flame retardant (V-0)

MATERIAL PROPERTIES — GPPS VS HIPS COMPARISON

PROPERTY	ASTM	GPPS	HIPS	UNITS
PHYSICAL / OPTICAL				
Specific Gravity	D792	1.05	1.04	—
Water Absorption, 24h	D570	0.01	0.05	%
Light Transmission	D1003	88	Opaque	%
MECHANICAL				
Tensile Strength, Ultimate	D638	6,500	3,500	psi
Tensile Modulus	D638	450,000	280,000	psi
Elongation at Break	D638	2	40	%
Flexural Strength	D790	10,000	5,500	psi
Flexural Modulus	D790	450,000	280,000	psi
Impact, Izod Notched	D256	0.3	2.5	ft-lb/in
Hardness, Rockwell	D785	M75	R85	—
THERMAL				
Heat Deflection @ 264 psi	D648	180	175	°F
Heat Deflection @ 66 psi	D648	200	195	°F
Max Continuous Service	—	160	150	°F
CTE (Linear)	D696	3.8 × 10 ⁻⁵	4.0 × 10 ⁻⁵	in/in/°F
Thermal Conductivity	C177	0.8	1.0	BTU-in/hr-ft²-°F
ELECTRICAL				
Volume Resistivity	D257	> 10 ¹⁶	> 10 ¹⁵	ohm-cm
Dielectric Strength	D149	500	350	V/mil
Dielectric Constant, 1MHz	D150	2.5	2.6	—
FLAMMABILITY				
UL Flammability Rating	UL 94	HB	HB	FR grades: V-0
DURABILITY				
Weathering / UV	G154	Poor	Poor	Yellows; UV-stab. avail.

PROCESSING / FABRICATION NOTES: Excellent machinability — scores & snaps cleanly (GPPS) | HIPS thermoforms easily at 300–375°F | Solvent-bondable with MEK or methylene chloride | Excellent paint & print adhesion | GPPS is brittle — avoid impact loads

CHEMICAL RESISTANCE: Good resistance to dilute acids, bases, alcohols, and aqueous solutions. **AVOID:** ketones, esters, aromatics, chlorinated solvents — will dissolve or craze polystyrene. Not suitable for outdoor use without UV stabilization.

Typical values; not to be used as specifications. Actual values may vary by grade, manufacturer, and processing conditions. Contact Plastic-Craft for grade-specific data.